

Eldridge Surianto

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

Bachelor of Science in Computer Engineering | GPA: 3.95

Expected Graduation: May 2027

- **Concentrations:** Systems and Architecture & Computing Hardware and Emerging Architectures
- **Related Coursework:** Programming for Hardware/Software Systems, Digital Design Laboratory, Data Structures & Algorithms, Design & Analysis of Algorithms, Objects & Design, Statistics & Applications

SKILLS

Programming: C/C++, Python, MATLAB, Verilog, VHDL, Rust, MIPS, Java, HTML, CSS, Javascript

Technical: FPGAs, Data Structures & Algorithms, Finite Element Analysis, Web Application Development, Machine Vision, Oscilloscopes

Tools: Vitis HLS, Vivado, Quartus Prime, Linux CLI, Django Web Framework, MATLAB Livelink, COMSOL Multiphysics

Platforms: Linux, Windows, MacOS

Languages: English, Chinese, German

PROFESSIONAL EXPERIENCE

Software Intern | Automation Technology Center PTE LTD | Singapore

May 2025 - August 2025

- Built a high-performance data processing pipeline, utilizing OpenCV for laser line image processing, OpenGL for interactive 3D visualization, and multithreading to reduce processing time for 500 images from over 60 seconds to less than 2 seconds; leveraged OpenMP for parallel computations, accelerating machine vision capabilities crucial for semiconductor manufacturing.
- Implemented a precision 3D laser scanning system for automated semiconductor lead frame inspection, enabling highly detailed and repeatable dimensional data capture by integrating EtherCAT (SOEM protocol) with a Yaskawa SGD7S servo drive for real-time encoder-triggered, low-level camera control via Spinnaker SDK acquisitions. The program has been successfully integrated into an industrial semiconductor device fabrication machine.

Undergraduate Research Assistant | Plasma & Dielectrics Lab | Georgia Tech

August 2024 - Present

- Designed MATLAB scripting pipelines for modeling zinc oxide voltage-dependent resistor (MOVs) microstructures in COMSOL Multiphysics and assigning semiconductor granular properties, reducing Finite Element Analysis (FEA) setup time by over 99%.
- Executed thousands of FEA simulations using the Monte Carlo method to analyze the impact of material inhomogeneity on current distribution, generating statistical insights to optimize MOV design by identifying key MOV parameters controlled during manufacturing process affecting performance for high-energy handling power electronics.

PROJECTS

HLS-Accelerated Box Blur with Rust Host FFI Integration

- Optimized Box Blur image filter algorithm using Vitis HLS for FPGA acceleration by iteratively optimizing its hardware implementation using techniques like on-chip line buffers and array partitioning, achieving latency reduction of 55%.
- Developed a Rust host application to control the HLS-accelerated filter kernel via Foreign Function Interface (FFI).

LLVM IR Analysis Tool

- Built a Linux command-line analysis tool using the llvm-ir crate to parse LLVM Intermediate Representation (IR), reporting per-function instruction counts, control flow structure, and categorized operation types (e.g., loads, stores, branches).
- Enabled low-level inspection of compiler-generated logic, aiding in debugging and understanding of HLS-like transformations, memory access patterns, and function-level performance cost in IR.

VHDL RTL LED Metronome on DE10-Standard ModelSim-Intel SoC FPGA

- Designed and implemented a VHDL-based LED metronome peripheral on a DE10-Standard FPGA using Quartus Prime and comprehensive ModelSim timing simulations, incorporating a state machine mechanism to manage the precise and visually engaging timing through PWM-controlled LED brightness and clock division for adjustable BPM.
- Integrated the peripheral into a custom SCOMP (Simple Computer) architecture, enabling dynamic tempo and LED control via SCOMP assembly instructions directly interfacing with memory-mapped registers.

LEADERSHIP EXPERIENCE

Product Owner | CS2340 Objects and Design | Georgia Tech

August 2024 - December 2024

- Led a 5-person team, strategically translating project requirements into actionable tasks and allocating responsibilities, while overseeing UI/UX and backend implementation.
- Applied SCRUM Agile principles to manage member progress and create full-stack Django web application projects.
- Guided the team to focus on user-centric features for two distinct applications: a Food Finder, leveraging Google Maps API for location-based search, user favorites, and history tracking (SQLite3 database); and a Spotify Wrapped app, utilizing Spotify Web API User Data to generate personalized listening habit visualization and AI-powered song recommendations (OpenAI LLM API).